Hospital Readmission Risk Analysis

Ruth Santiago BBA
Financial Data Analyst | People Analytics | Mortgage Risk & Credit Expertise | SQL | Excel | Tableau | AI Tools | Turning Data into Strategic Decisions

March 12, 2026

Project Overview

Unplanned hospital readmissions signal potential gaps in care — whether in recovery, discharge planning, or follow-up. They also carry significant cost implications for healthcare systems, making them a persistent focus for hospital leadership and quality improvement teams.

This project analyzes **101,766 patient encounters** from the *Diabetes 130-US Hospitals dataset* to examine how **hospital stay duration, diagnostic intensity, patient complexity, and medication adjustments** relate to readmission outcomes. SQL was used for data exploration and preparation, while Tableau was used to build the visualizations.

Readmission outcomes are grouped into three categories:

- Readmitted within 30 days
- Readmitted after 30 days
- No readmission

This structure allows the analysis to distinguish between **early readmissions — which often reflect acute complications — and later returns, which may indicate chronic disease management challenges.**

Data Description

The dataset covers inpatient encounters across 130 U.S. hospitals for patients with a diabetes diagnosis. Each record represents a single visit and includes:

- Number of diagnoses associated with the patient
- Length of hospital stay (in days)
- Number of laboratory procedures performed
- Medication changes made during hospitalization
- Readmission status: within 30 days, after 30 days, or none

Together, these variables support analysis of how treatment intensity and patient complexity influence hospital utilization and return visits.

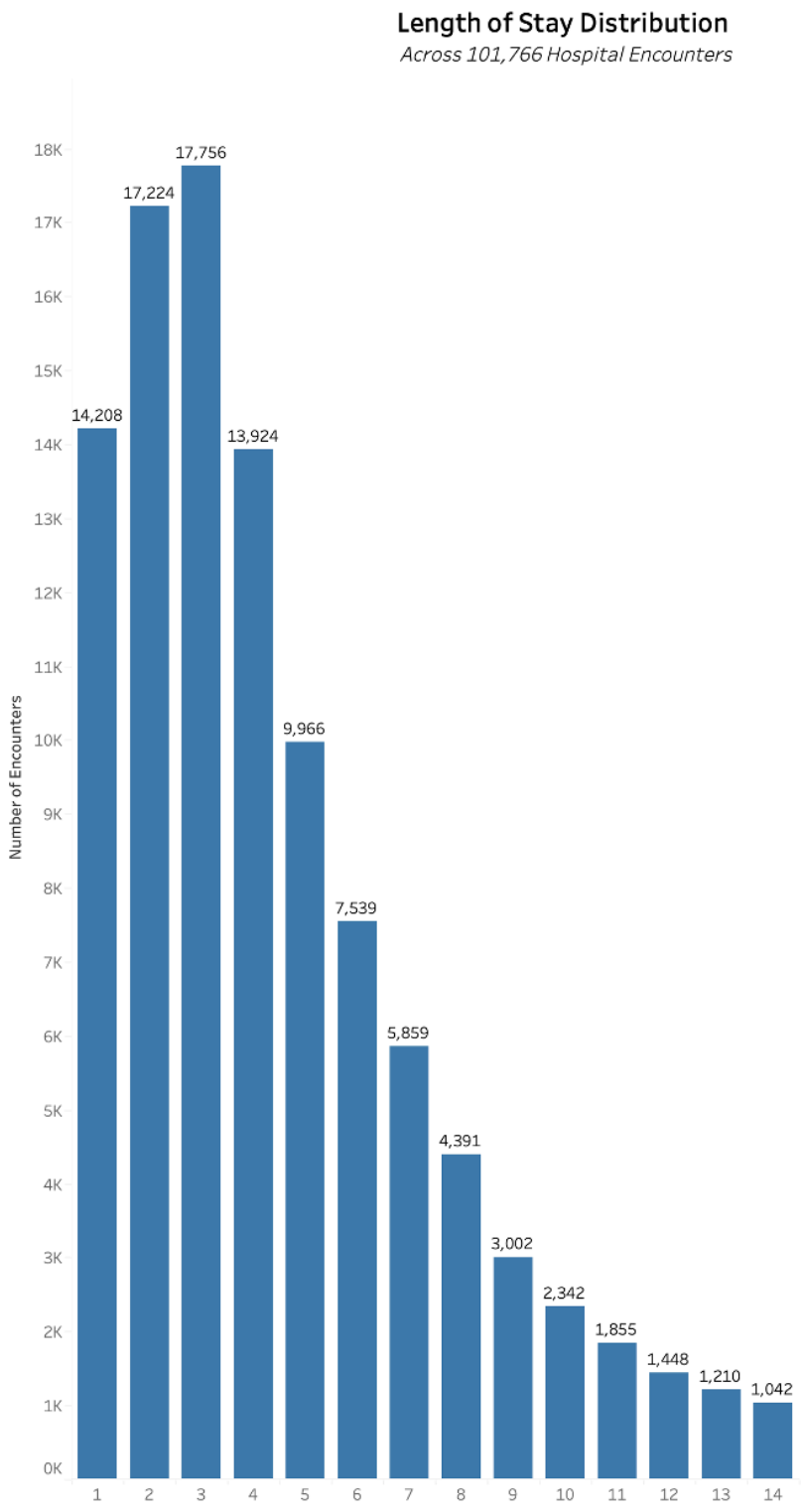
Hospital stays in this dataset are concentrated in the first few days of admission. The highest frequency occurs at three days (17,756 encounters), and approximately 47% of all encounters last between 2 and 4 days. Stays longer than 7 days account for less than 15% of total encounters.

After day five, encounter counts decline steadily, indicating that most patients are discharged within the first week. Extended stays beyond seven days become progressively less common, with only a small share of encounters reaching past ten days — a pattern consistent with typical hospital operations where many admissions involve short-term observation or routine treatment.

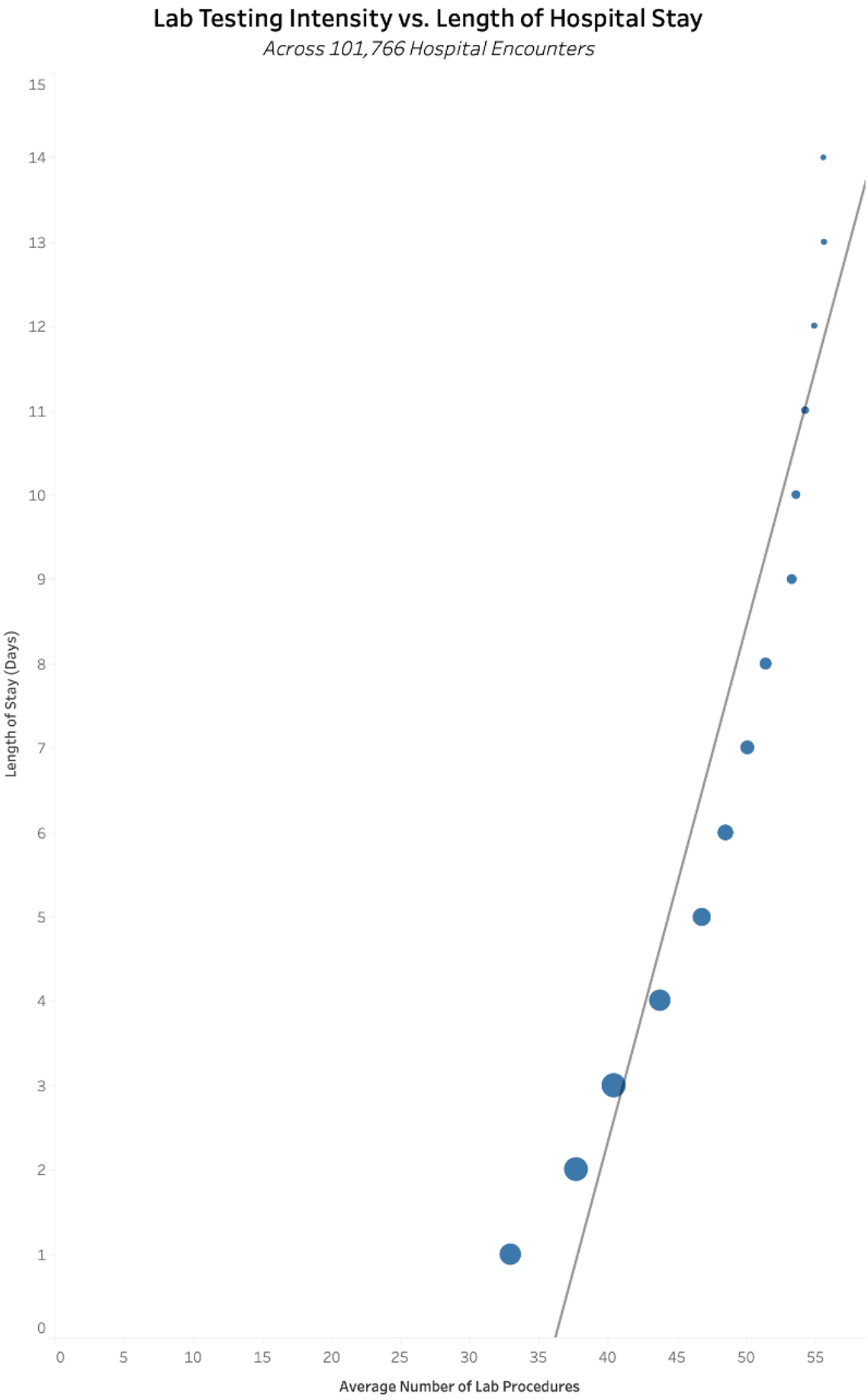
KEY INSIGHT

Short stays dominate, but the long-stay tail is clinically meaningful. Encounters exceeding 7 days likely represent the most complex cases in the dataset — patients who may warrant closer monitoring and

more intensive discharge planning.



Most hospital stays peak between 2 and 3 days, with encounter counts declining steadily after day 5, suggesting most patients are discharged within the first week.



Lab testing intensity increases as hospital stays become longer, suggesting that extended hospitalizations are typically associated with more complex cases requiring additional diagnostic monitoring.

To explore the relationship between clinical activity and hospitalization duration, I analyzed how the average number of laboratory procedures changes as patient length of stay increases across 101,766 hospital encounters.

The results show a strong positive relationship between the number of laboratory tests performed and the length of hospital stay. Patients with shorter hospitalizations generally receive fewer laboratory procedures, while patients with longer stays tend to undergo more extensive diagnostic testing.

Encounters lasting 1 to 3 days typically involve approximately 34–41 laboratory procedures on average, while hospital stays of 7 days or longer are associated with 50 or more lab procedures. For the longest stays observed in the dataset (around 13–14 days), the average number of laboratory procedures exceeds 55 tests per encounter.

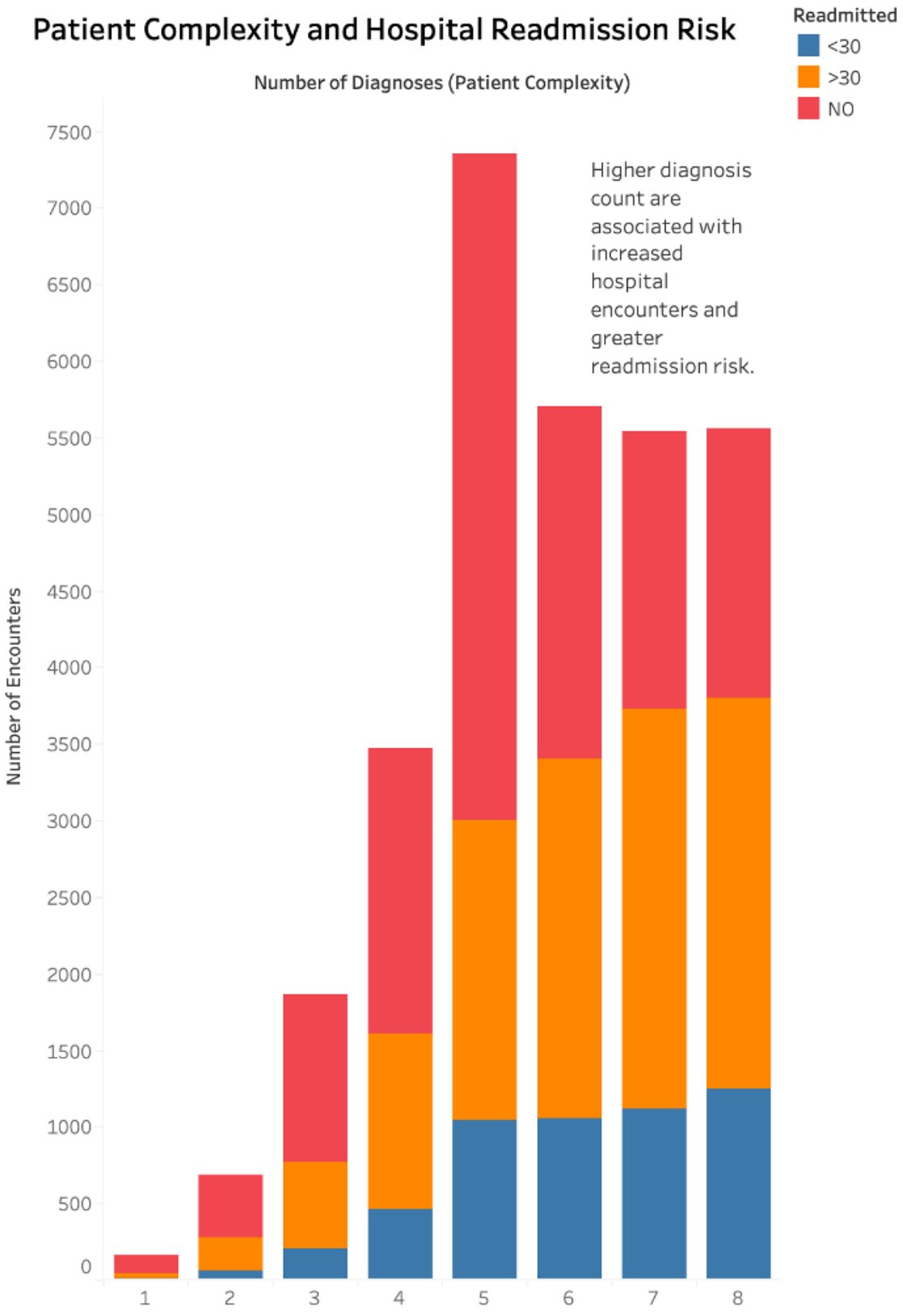
This trend suggests that longer hospitalizations are often accompanied by increased diagnostic monitoring and clinical evaluation, which may reflect more complex medical conditions, complications during treatment, or the need for continuous patient monitoring.

The upward trend line in the chart reinforces this pattern, indicating that lab testing intensity increases consistently as length of stay grows. This relationship highlights how diagnostic activity tends to scale with the complexity and duration of patient care.

Understanding this relationship can provide valuable insights for hospital operational planning and resource allocation, as longer hospital stays may require greater laboratory capacity, staffing, and diagnostic resources.

KEY INSIGHT

Lab volume is a proxy for clinical complexity. Patients accumulating high lab counts early in a stay may be on a trajectory toward longer hospitalization — an early-warning pattern worth building into discharge planning workflows.



Patient complexity is measured by the number of diagnoses recorded per hospital encounter.

To explore how patient complexity relates to hospital utilization, I analyzed the relationship between the **number of diagnoses recorded during a hospital encounter and patterns of hospital encounters and readmission risk**.

The chart shows that as the **number of diagnoses increases**, both the total number of hospital encounters and the likelihood of readmission tend to rise. Patients with **fewer diagnoses** represent a smaller portion of hospital activity, while patients with **multiple diagnoses** account for a larger share of encounters.

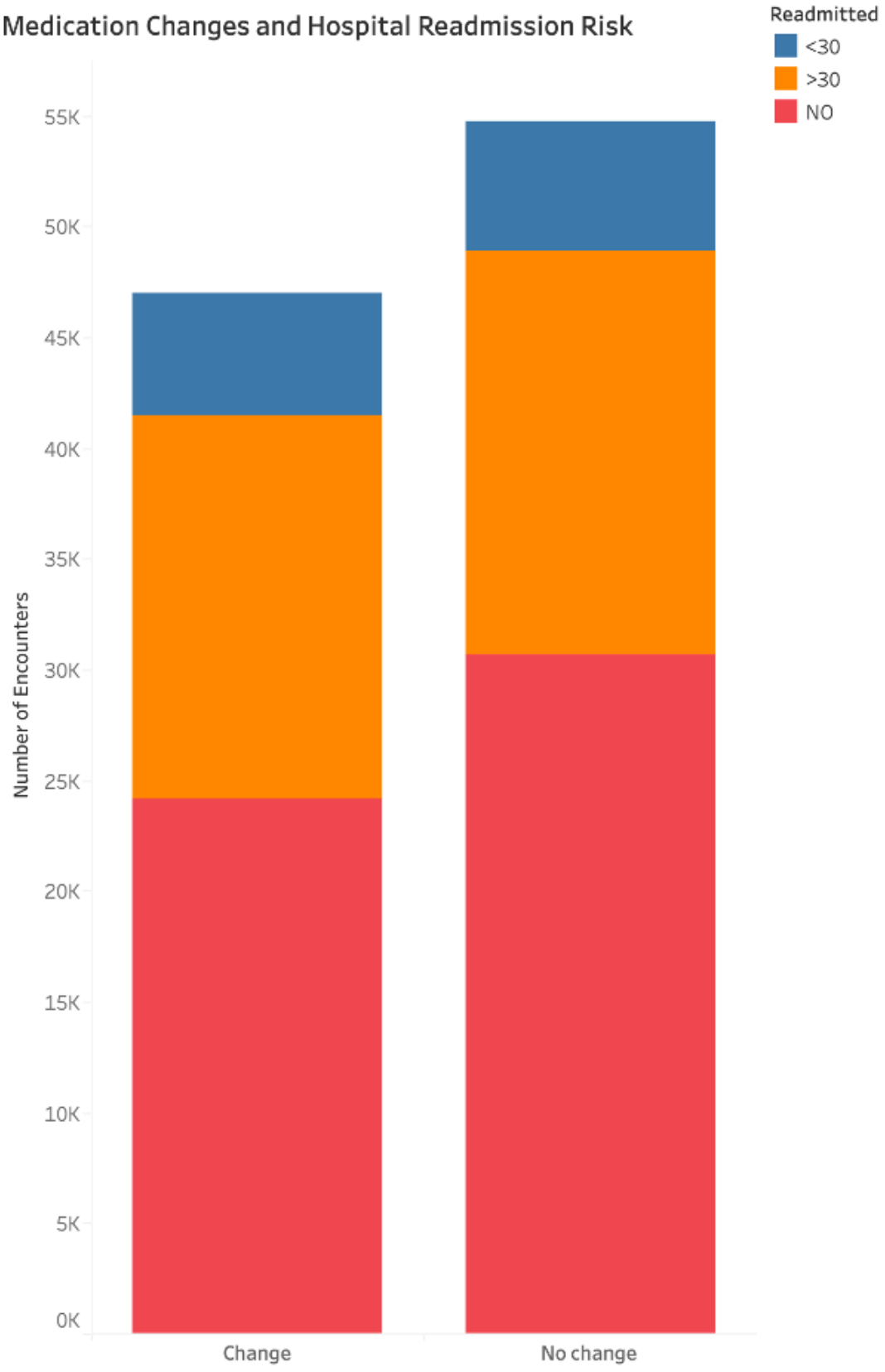
Encounters involving **five or more diagnoses** show a noticeable increase in hospital activity and readmission-related patterns. This suggests that patients with multiple coexisting conditions may require more frequent monitoring, follow-up care, and additional medical interventions.

Higher diagnosis counts often reflect **greater patient complexity**, which may include chronic illnesses, comorbidities, or conditions requiring coordinated treatment across multiple specialties. As clinical complexity increases, the likelihood of complications, additional testing, and potential readmissions may also rise.

Understanding the relationship between patient complexity and hospital utilization can help healthcare organizations **identify high-risk patients earlier, allocate clinical resources more effectively, and develop targeted care management strategies aimed at reducing avoidable readmissions**.

KEY INSIGHT

Diagnosis count is one of the strongest complexity signals in this dataset. Patients with five or more diagnoses are not just sicker — they are more likely to return. Targeting this group for enhanced discharge support and post-hospital follow-up represents a concrete, data-supported intervention point.



Medication change status indicates whether any medication adjustments were recorded during the hospital encounter.

To examine whether treatment adjustments during hospitalization relate to readmission outcomes, I analyzed encounters based on whether **medication changes were recorded during the hospital stay**.

The chart compares encounters in which medications were **adjusted** with those where **no medication changes occurred**, along with the associated readmission outcomes. Encounters with **no medication changes** represent the largest portion of hospital visits in the dataset. However, the distribution suggests that encounters involving **medication adjustments** may be associated with a slightly lower proportion of readmissions compared with those where medication regimens remained unchanged.

Medication changes during hospitalization often reflect **active clinical management**, where physicians adjust treatment plans in response to diagnostic findings, patient symptoms, or treatment effectiveness. These adjustments may help stabilize patients before discharge and improve overall care coordination.

Although this analysis does not establish a causal relationship, it highlights how **treatment decisions made during hospitalization may influence patient outcomes after discharge**. Monitoring how medication management interacts with readmission risk can help healthcare organizations identify opportunities for improving discharge planning and post-hospitalization care.

Understanding these patterns may support **better care management strategies, improved medication reconciliation processes, and targeted interventions aimed at reducing avoidable readmissions.**

KEY INSIGHT

Active medication management appears to be associated with modestly better readmission outcomes. While this analysis does not establish causation, it points to medication reconciliation and treatment adjustment as levers worth examining in discharge planning and post-hospitalization care programs.

Key Findings

Taken together, the four analyses point to a consistent pattern: readmission risk is not driven by any single factor, but by the accumulation of complexity indicators across a patient's encounter.

Most hospital stays are short, but the long-stay population is disproportionately complex. Nearly half of encounters last 2–4 days. Encounters exceeding 7 days are fewer in number but likely represent the highest-acuity cases.

Diagnostic intensity tracks with hospital stay length. Higher lab procedure counts correspond with longer stays, pointing to patients who require sustained clinical evaluation and monitoring.

Patient complexity — measured by diagnosis count — is the strongest readmission signal. Patients with five or more diagnoses show higher encounter frequency and greater readmission likelihood, making them a clear target for enhanced care coordination.

Active medication management is associated with modestly lower readmission rates. Encounters with medication adjustments show a slightly better readmission profile than those without, suggesting treatment responsiveness during hospitalization may matter for post-discharge outcomes.

Recommendations for Hospital Staff

The patterns in this analysis point to three areas where targeted action may reduce avoidable readmissions.

Prioritize High-Complexity Patients for Enhanced Discharge Support

Patients with five or more diagnoses represent a well-defined, data-identifiable group with elevated readmission risk. Hospitals can reduce return visits for this population by scheduling structured post-discharge check-ins within 7 days, routing complex patients to care coordinators before discharge rather than after readmission, and using telehealth touchpoints to monitor recovery without requiring in-person visits.

Build Early-Warning Triggers Around Lab Volume and Stay Length

Since high lab counts and extended stays are associated with greater complexity and readmission risk, hospitals can operationalize these signals by flagging encounters where lab procedure counts exceed a threshold by day two or three, and routing flagged patients to discharge planning teams earlier in the hospitalization.

Strengthen Medication Reconciliation at Discharge

Encounters with medication adjustments show modestly better readmission outcomes, which points to active treatment management as a protective factor. Hospitals can build on this by providing clear written summaries of any medication changes at discharge, conducting pharmacist-led reviews for patients with multiple adjustments, and ensuring follow-up appointments are scheduled before discharge.

Conclusion

This analysis demonstrates that hospital readmission risk in a diabetic patient population is most clearly associated with patient complexity — particularly diagnosis count and diagnostic intensity — rather than any single treatment variable. The finding that active medication management is associated with lower readmission rates adds a practical dimension: treatment decisions made during the hospitalization itself may influence whether patients return.

The practical implication is direct: hospitals do not need to wait for a readmission to identify high-risk patients. Complexity indicators available at or shortly after admission can support earlier, more proactive intervention. Strengthening discharge planning, improving post-hospital follow-up, and building care coordination around the highest-complexity patients represent the clearest opportunities for reducing avoidable readmissions.

Data does not prevent readmissions — but it can help hospitals direct their attention and resources to the patients most likely to need them.

Thank you for taking the time to review this project. If you have any questions, feedback, or would like to discuss this analysis further, please feel free to reach out.

I am always open to ***collaboration, learning opportunities, and conversations about how data can support better decision-making****. I look forward to connecting with others in the data and analytics community and exploring future opportunities.*

#DataAnalytics #SQL #HealthcareAnalytics #DataVisualization #Tableau #DataProjects #LearningInPublic #AnalyticsPortfolio